

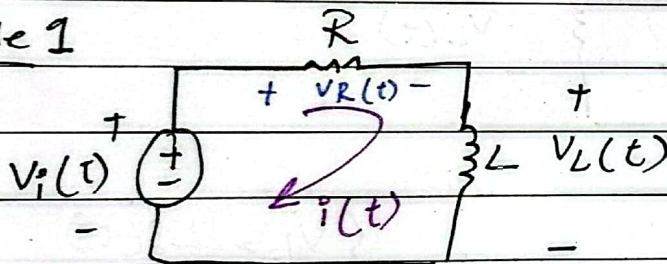
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State Space Representation (electrical)

- 1/ The no. of state variable and hence the state equation = The number of reactive element in the circuit (i.e. L & C)
- 2/ All state eqs & output equations must be written in terms of all state variable and input variable
- 3/ For capacitor $\rightarrow V_C$ (voltage across capacitor) is taken as state variable
 For inductor $\rightarrow I_L$ (current through inductor) is taken as state variable

Example 1



1 reactive element
 1 state variable
 1×1 A matrix

$\Rightarrow 'L' \Rightarrow x(t) = i(t)$ (Reactive element)
 $u(t) = V_i(t)$ (Input)
 $y(t) = V_R(t)$ (Output)

applying KVL $\rightarrow L \frac{di}{dt}$ (voltage across inductor)

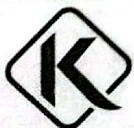
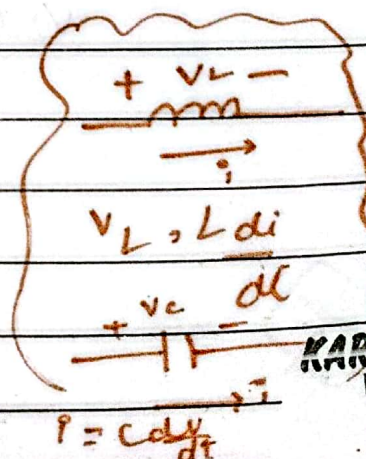
$$-V_i(t) + V_R(t) + V_L(t) = 0$$

$$V_R = R i(t) + L \dot{i}(t) = V_i(t)$$

$$L \dot{i}(t) = V_i(t) - R i(t)$$

$$\dot{i}(t) = \frac{V_i(t) - R i(t)}{L}$$

$$V_i = V_R + V_L$$



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$$\dot{i}(t) = \left[-\frac{R}{L} \right] i(t) + \left[\frac{1}{L} \right] v_i(t)$$

$$\dot{x}(t) = A x(t) + B u(t)$$

$$y(t) = i(t) + v_L(t)$$

Now across output

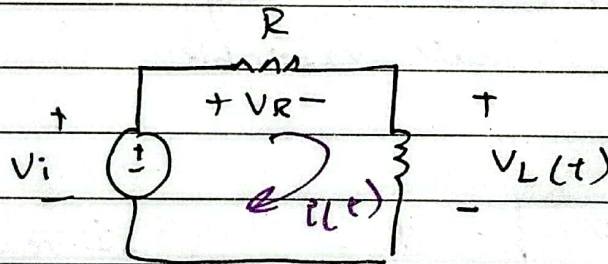
$$y = v_R(t) = R i(t)$$

$$y = \begin{bmatrix} R \end{bmatrix} i(t) + \begin{bmatrix} 0 \end{bmatrix} v_i$$

$$y = C x(t) + D u(t)$$

$$A = \begin{bmatrix} -\frac{R}{L} \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{L} \end{bmatrix} \quad C = \begin{bmatrix} R \end{bmatrix} \quad D = 0$$

Example 2



$$'L' \Rightarrow x(t) = i(t)$$

$$u(t) = v_i(t)$$

$$y(t) = v_L(t)$$

$$v_L = L \dot{i}(t)$$

$$\frac{v_L}{L} = \dot{i}(t)$$

$$-v_i(t) + v_R + v_L(t) = 0$$

$$v_L(t) = v_i(t) - v_R$$

$$v_L(t) = v_i(t) - i(t)R$$

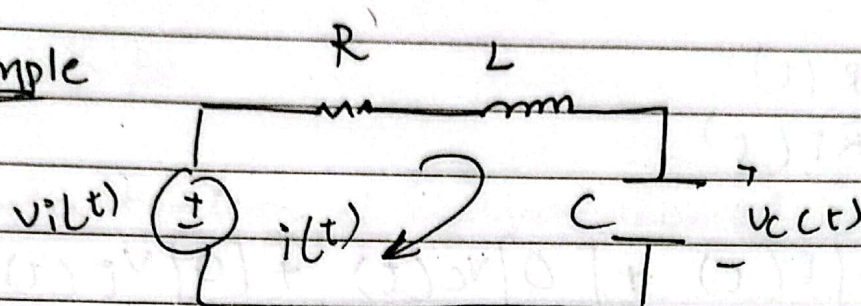
$$v_L(t) = -R i(t) + v_i(t)$$

Substitute $\dot{i}(t)$ here

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Example



2 reactive element

2 state variable

State variable $i(t), v_c(t)$

$$\dot{i}(t) = a_1 i(t) + b_1 v_c(t) + d_1 v_i(t)$$

$$\dot{v}_c(t) = a_2 i(t) + b_2 v_c(t) + d_2 v_i(t)$$

$$x(t) = \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix}$$

$$y(t) = v_R(t)$$

$$-v_i(t) + v_R(t) + L\dot{i}(t) + v_c(t) = 0$$

$$-v_i(t) + i(t)R + L\dot{i}(t) + \frac{1}{C} \int i(t) dt = 0 \quad \text{VC which is state variable}$$

$$Ri(t) + L\dot{i}(t) + v_c = v_i(t)$$

$$L\dot{i}(t) = v_i(t) - Ri(t) - v_c$$

$$\dot{i}(t) = \left[-\frac{R}{L} \right] i(t) - \frac{1}{L} v_c(t) + \frac{1}{L} v_i(t)$$

$$i(t) = C \frac{dv_c}{dt}$$

$$v_c(t) = C \dot{v}_c(t)$$

$$\dot{v}_c(t) = \frac{1}{C} i(t) + b(v_c(t)) + (0) v_i(t)$$

$$\begin{bmatrix} \dot{i}(t) \\ \dot{v}_c(t) \end{bmatrix} = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix} \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} v_i(t)$$



$\dot{i}(t)$ $\xrightarrow{A}$ x $\xrightarrow{B}$ U

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$$y = v_R(t)$$

$$y = Ri(t)$$

$$y = [R]i(t) + [0]v_C(t) + [0]v_L(t)$$

$$y = \underbrace{[R \quad 0]}_C \underbrace{\begin{bmatrix} i(t) \\ v_C(t) \end{bmatrix}}_x + \underbrace{[0]}_D \underbrace{v_i(t)}_u$$

$$y(t) = \begin{bmatrix} v_R(t) \\ v_L(t) \end{bmatrix}$$

$$v_R(t) = Ri(t) + (0)v_C + (0)v_L$$

$$v_L(t) = v_i(t) - v_R - v_C$$

$$= -Ri(t) - v_C + v_i(t)$$

$$\begin{bmatrix} v_R \\ v_L \end{bmatrix} = y = \begin{bmatrix} R & 0 \\ -R & -1 \end{bmatrix} \begin{bmatrix} i(t) \\ v_C(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} v_i$$

